SPECIAL SERIES: CARING FOR OLDER ADULTS WITH CANCER

# Sarcopenia in the Older Adult With Cancer

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## INTRODUCTION

The term sarcopenia, derived from the Greek words *sarx* for flesh and *penia* for loss, was first introduced by Rosenberg[1] in 1989 to describe the loss of muscle mass that accompanies aging. As he describes, there is probably no decline in structure and function more dramatic than the decline in lean body mass or muscle mass over the decades of life.[1] Beginning as early as the fourth decade of life, there is a linear decrease in muscle mass and strength that accompanies increasing age.[2] This age-related decline has been the focus of gerontology research and the basis for understanding functional decline and disability for several decades. Sarcopenia in the older adult has been associated with functional impairments, disability, increased risk of falls and fractures, reduced health-related quality of life (HRQOL), and increased risk of death.[2,3] As sarcopenia often results in disability with a loss in independence necessitating caregiver assistance and/or long-term care placement, it has been estimated to cost the US health system approximately $18.5 billion in US dollars per year.[4]

While muscle strength and mass decrease with age, body fat gradually increases up until the seventh decade of life.[5] Despite declining resting metabolic rates and physical activity levels with age, food intake remains disproportionately high, leading to yearly positive energy balance and weight gain.[6] Thus, the development of sarcopenia is often accompanied concurrently with a gradual increase in adiposity termed sarcopenic obesity. Given the rising rates of obesity in the United States and abroad,[7] the prevalence of this debilitating geriatric syndrome is increasing with a synergistic risk from both sarcopenia and obesity.[8]

More recently, sarcopenia is of increasing interest in oncology because of its high prevalence and association with adverse outcomes.[9-12] Although the use of computed tomography (CT) for analysis of body composition was pioneered by Steven Heymsfield as early as the 1980s, it was not until 2004 that Shen et al described the methodology for estimating total-body skeletal muscle and adipose tissue from a single abdominal CT cross-sectional image.[13-15] This method was later adapted specifically for the use of quantifying body composition in oncology research using archived images acquired during routine care.[16] Only a few short years later in 2007, Prado et al[17] used these methodologies to describe body composition as an independent determinant of fluorouracil based chemotherapy toxicities. From these initial studies, there has been a flurry of publications across nearly every cancer type and stage examining the association of CT-based body composition metrics and cancer outcomes. As of February 2021, a PubMed search using the terms sarcopenia and cancer resulted in 1,821 hits. Although sarcopenia has been a focus of research for many years, no broadly accepted consensus definition of sarcopenia exists, in oncology or elsewhere. The definition of sarcopenia is probably one of the least agreed upon and most debated topics in body composition research.[18] Sarcopenia was first defined as two standard deviations below the mean muscle mass of healthy younger adults by Baumgartner et al[19] using dual-energy X-ray absorptiometry (DEXA). More recently, the definition of sarcopenia has broadened to not only involve low muscle mass from one or more methods, but also reduced strength and/or physical performance.[20] Although there is consensus within the larger gerontology literature that the definition of sarcopenia should include muscle strength and function as well as muscle mass, cancer-associated sarcopenia research has relied more heavily on CT-determined muscle mass as a diagnostic criterion. Complicating this further, a variety of CT derived cut-points have been used to characterize what is normal muscle mass in patients with cancer.[12] Part of the discrepancy in the definition of sarcopenia between the gerontologic and oncologic literature exists because of the different viewpoints and outcomes under consideration. Geriatrics views sarcopenia in the light of developing disability, whereas oncology has focused more on low muscle mass and its association with increased mortality and complications from cancer treatments. Regardless, the lack of a consensus definition of sarcopenia has limited the generalizability and comparison of study findings and overall progress in the field.

Given the variety of definitions applied, the prevalence of sarcopenia varies widely from 5%-89% based on the definition or cut-point applied and the population of interest.[9,12] In recent reviews of the literature, 11 different definitions of sarcopenia using total abdominal muscle area were used to calculate prevalence and evaluate prognosis across the included studies (Table 1).[9,12] As illustrated in Table 1, summary of the early proposed definitions of sarcopenia in oncology, the identified cut-point varies depending on

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## CONTEXT

**Key Objective**

What is sarcopenia and why is it important in older adults with cancer?

**Knowledge Generated**

Older adults with cancer are at increased risk of sarcopenia, and in turn, increased risk of chemotherapy toxicities, tumor progression, and mortality. Sarcopenia can be assessed using a variety of available methods and individuals with sarcopenia may benefit from targeted interventions consisting of exercise and/or nutrition.

**Relevance**

Detecting sarcopenia in the growing number of older adults with cancer can aid in the assessment of the risk and benefit ratio of cancer treatments and assist in targeting interventions. Still, many important knowledge gaps persist in our understanding of sarcopenia in oncology and more research is needed.

the cancer population as well as the geographic area. Of particular interest is that while the prevalence of sarcopenia is higher in older adults, the phenomenon of low muscle mass is not limited to the older adult population in oncology. For example, in a recent study of more than 3,000 adults with early stage colorectal cancer, 58.3% of adults older than 70 years and 26.8% of adults younger than 60 years suffered from sarcopenia.[21]

## CAUSES OF SARCOPENIA

Sarcopenia can arise and progress because of normal physiologic aging and inactivity or, secondarily, as a consequence of chronic disease states like chronic obstructive pulmonary disease, AIDS, or cancer.[1] Decreases in anabolic hormones, particularly testosterone, contribute significantly to age-related sarcopenia.[22] Structurally, age-related sarcopenia is characterized by a decrease in the size of type II muscle fibers (also known as fast-twitch muscle fibers).[23] Type II fiber atrophy is driven by several physiologic processes of aging, including loss of motor neurons, muscle denervation, neuromuscular junction instability, and decreases in muscle satellite cells responsible for growth, maintenance and repair of muscle.[23,24] Aging also results in mitochondrial dysfunction characterized by reduced mitochondrial DNA and ATP production in skeletal muscle and the accumulation of intracellular reactive oxygen

**TABLE 1.** Chronologic List of the Early Proposed CT-Based Definitions for Identifying Low Muscle Mass From Total Abdominal Muscle Area in Oncology and Their Associated Prevalence Estimates[12]

| | | Definition of Sarcopenia | | |
|---|---|---|---|---|
| **Study** | **Cancer Population** | **Males ($cm^2/m^2$)** | **Females ($cm^2/m^2$)** | **Sarcopenia Prevalence (%)** |
| Non-Asian populations[a] | | | | |
| Prado et al[25] | Obese patients with solid tumors of respiratory or GI tract, N = 250 | < 52.4 | < 38.6 | 15 |
| Baracos et al[26] | Non–small-cell lung cancer, N = 441 | < 55.4 | < 38.9 | 47 |
| van Vledder et al[27] | Colorectal cancer undergoing hepatic resection, N = 196 | < 43.8 | < 41.1 | 19 |
| Martin et al[28] | Lung or GI cancers, N = 1,473 | BMI < 25 $kg/m^2$: ≤ 43.0<br>BMI ≥ 25 $kg/m^2$: ≤ 53.0 | < 41 | NR |
| Camus et al[29] | Older adults with diffuse large B-cell lymphoma, N = 80 | < 55.8 | < 38.9 | 55 |
| Asian populations | | | | |
| Fujiwara et al[30] | Hepatocellular carcinoma, N = 1,257 | < 36.2 | < 29.6 | 11 |
| Iritani et al[31] | Hepatocellular carcinoma, N = 217 | < 36 | < 29 | 11 |
| Nakamura et al[32] | Diffuse large B-cell lymphoma, N = 207 | < 47.1 | < 34.4 | 56 |
| Choi et al[33] | Pancreatic cancer, N = 484 | < 42.2 | < 33.9 | 21 |
| Harada et al[34] | Esophageal cancer, N = 325 | < 44.5 | < 36.5 | 33 |
| Kimura et al[35] | Non–small-cell lung cancer, N = 134 | < 41.0 | < 38 | 46 |

Abbreviations: BMI, body mass index; CT, computed tomography; NR, not reported.

[a]CT-based definitions are grouped by geographic area (non-Asian *v* Asian) to highlight geographic differences in proposed definitions.

species.[36] Muscle fiber atrophy is accompanied by the infiltration of adipose tissue, which produces a decrease in muscle quality.[37]

Cancer and its treatment can increase the risk of developing sarcopenia and exacerbate pre-existing muscle wasting in older adults.[38] Cancer can potentiate many of the factors that contribute to age-related sarcopenia including anorexia, inactivity, and a proinflammatory state (Fig 1). In addition, many cancer treatments, particularly chemotherapeutic agents, can cause both indirect (anorexia, nausea, and fatigue) and direct damage to muscle tissue by pathways that upregulate proteasome activity, activate mitogen-activated protein kinase and extracellular regulated kinase signaling, and induce mitochondrial dysfunction irrespective of anorexia or nutrition.[39-41]

Sarcopenia has significant clinical overlap with cachexia, although the two are distinct wasting syndromes.[42] Cachexia is diagnosed exclusively in disease states like cancer and is characterized as a syndrome of both weight and muscle loss with or without loss of fat mass (Fig 2).[43] Like sarcopenia, cachexia is quite common in older adults with cancer and is accompanied by decreased muscle function and reduced physical performance.[42,43] Although several physiologic derangements like low testosterone and mitochondrial dysfunction occur in both conditions, their pathophysiologies differ. Cachexia is driven by a procatabolic state that is influenced by tumor metabolism and by proinflammatory cytokines (ie, interleukin-6 and tumor necrosis factor) that are directly secreted by tumor tissue or produced by the host's immune response.[44] These factors lead not only to protein and muscle degradation, but also to lipolysis.

Frailty also has a close relationship and overlap with sarcopenia, but the two terms are certainly not interchangeable.[45] Frailty is rarer and represents a state in which older adults are vulnerable under stress.[46] A critical hallmark shared by sarcopenia and frailty is reduction in physical function; both conditions can be accompanied by a change in mass, although in frailty, it could be fat or lean mass.[47] The frailty phenotype (three of five of the following: weakness, slowness, weight loss, fatigue, and decreased activity) described by Fried et al,[47] however, encompasses a broader condition that can be influenced by other geriatric syndromes including sarcopenia, neurocognitive impairments, mood disorders, and socioeconomic factors. Frailty, sarcopenia, and cachexia have important areas of overlap and distinct differences with each other, with entity-associated adverse outcomes and vulnerability. Table 2 compares the common diagnostic criteria for sarcopenia, cancer cachexia, and frailty.[20,43,47]

## METHODS FOR IDENTIFYING SARCOPENIA

The growth and emergence of new imaging techniques have greatly affected the field of body composition research. Although body weight and body mass index are well-known and commonly used measures in clinical practice to gauge health and nutritional status, they are inaccurate, particularly in older patients, and unable to differentiate between various tissue compartments.[18,48] Today, many different techniques and methodologies are available that can be used in the assessment of body composition ranging from simple anthropometric measures to more advanced magnetic resonance imaging (MRI) and even patient-reported measures. See Table 3 for a list of commonly used body composition techniques.

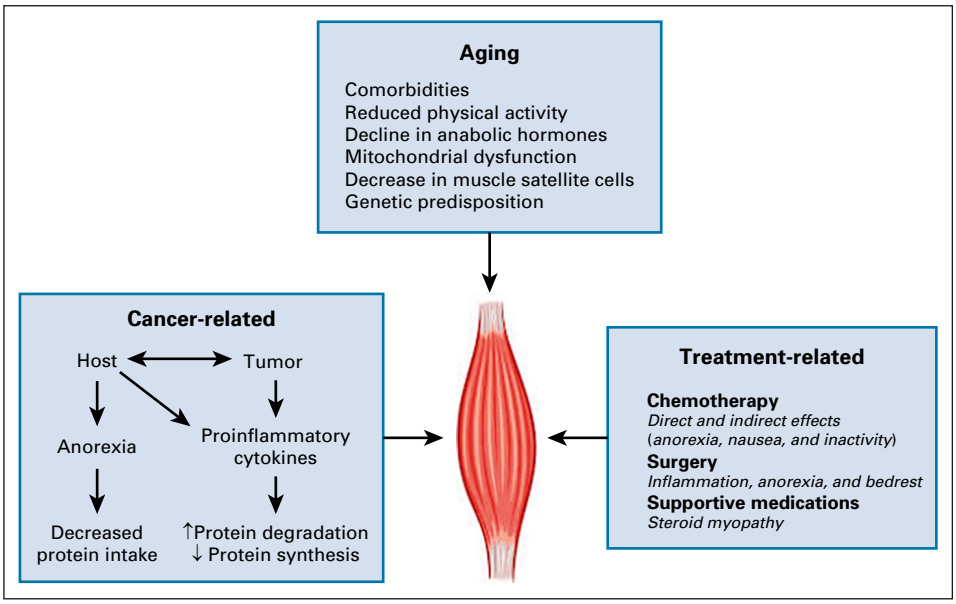


**FIG 1.** Multifactorial causes of muscle loss in the older adult with cancer. ↑, increase; ↓, decrease.

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| Features | Sarcopenia | Frailty | Cachexia |
|---|---|---|---|
| Muscle mass | **↓** | ↓ or ↔ | ↓ |
| Muscle strength and function | **↓** | **↓** or ↔ | ↓ |
| Fat mass | ↔ | ↔ | ↓ |
| Basal metabolic rate | ↓ | ↓ | ↑ |
| Inflammation | ↔ | ↑ | ↑ |
| Overall body weight | ↔ | **↓** | **↓** |

**FIG 2.** Comparisons of the similarities and distinctions between sarcopenia, cachexia, and frailty. The bolded arrows indicate these are part of the formal definition of that entity. ↑, increase; ↓, decrease; ↔, can be increased, decreased, or unchanged with no real effect.

One of the most commonly used imaging techniques within the nononcologic literature is DEXA.[49] DEXA imaging is a very precise method for quantifying whole-body and regional composition with minimal radiation exposure. However, DEXA provides only an estimate of lean body mass or fat-free mass and is typically unable to decipher total body skeletal muscle mass nor differentiate types of adipose tissue.[18] In addition, there is a variety of different hardware and software packages across manufacturers resulting in inconsistencies between machines, which limits comparability.

Within oncology, muscle mass is most often quantified by routine CT imaging, given their common use for cancer staging, disease monitoring, and cancer surveillance. Although CT is more costly and complex compared with many other measures, its routine use within many cancer types in oncology makes it very attractive for wide-scale use. Practical and precise measurement approaches have been developed to quantify body composition using CT images acquired during routine care.[14,16] This approach commonly involves the use of a single cross-sectional slice to estimate whole-body muscle volume. The third lumbar vertebra landmark has been identified as the strongest correlate of total adiposity and muscle or fat-free mass when compared with MRI and DEXA.[14,16] Given the abundance of CT imaging within oncology, the majority of the sarcopenia literature in cancer is derived from archived CT images. One additional advantage of using CT imaging is the ability to provide additional qualitative measurements of muscle and fat.[50] Muscle radiodensity provides information regarding the composition of muscle otherwise only available from muscle biopsy. Attenuation of muscle is inversely related to muscle fat content and can be used as a surrogate measure of muscle quality.[50] Fat infiltration in skeletal muscle, known as myosteatosis, is represented by lower muscle density and has been shown to correlate with frailty and physical functional impairments in older adults with cancer.[45,51]

Bioelectrical impedance analysis (BIA) is a low-cost, quick, and safe tool to estimate fat-free mass that can be performed at the bedside. BIA uses a small electrical signal sent through the body and measures resistance to provide estimations of total body water and fat-free mass.[18] In a recent systematic review of 24 studies using BIA for the identification of sarcopenia in cancer, BIA was an accurate method for detecting sarcopenia and for evaluating associations with adverse outcomes.[52] The primary drawback and barrier to the use of BIA in oncology is the lack of precision with fluctuations in hydration status and in the presence of edema.[52] Nevertheless, if used carefully and in a standardized fashion, BIA represents an inexpensive and simple tool that does not require skilled personnel.

More novel and/or accurate techniques of assessing skeletal muscle mass have been developed over the past few years. Most notably, the use of the $D_3$-creatine dilution method is a direct assessment of muscle mass that is gaining traction within the gerontologic literature.[53,54] Although a promising method, $D_3$-creatine dilution has not been reported on within oncology and warrants further use and validation at this time. In addition, patient-reported measures that ascertain impairments in strength and functioning are becoming increasingly common. In a recent study of strength, assistance with walking, rising from a chair, climbing stairs, and falls (SARC-F), a five-item questionnaire for rapid screening of sarcopenia, 33% of older adults with cancer screened positive, and positivity

**TABLE 2.** The Diagnostic Criteria of Sarcopenia From the European Working Group on Sarcopenia in Older People Compared With the Diagnostic Criteria of Cancer Cachexia Based on an International Consensus Definition and the Fried Frailty Phenotype

| Parameter | Sarcopenia[20] | Cancer Cachexia[43] | Frailty[47] |
|---|---|---|---|
| Diagnostic criteria | Low muscle strength<br>Low muscle quantity[a] or quality<br>Reduced physical performance | Weight loss > 5% over the past 6 months or<br>2% weight loss and either<br>BMI < 20 or<br>Evidence of muscle depletion | Weight loss or low muscle mass<br>Weakness<br>Exhaustion or poor endurance<br>Slowness<br>Low activity |
| Definition | Criteria 1 = probable sarcopenia<br>Criteria 1 + 2 = sarcopenia<br>Criteria 1 − 3 = severe sarcopenia | Any of the above scenarios constitutes cachexia | 1 or 2 criteria = intermediate or prefrail<br>3 or more = frailty |

Abbreviation: BMI, body mass index.

[a]Low muscle mass quantity alone is the most commonly used sarcopenia definition within the oncology literature.

**TABLE 3.** Common Techniques for Identifying Sarcopenia

| Technique | Pros | Cons | Common Definitions[a] |
|---|---|---|---|
| Anthropometry | Easily obtained and inexpensive | Relatively insensitive and significant interobserver variability | Calf circumference < 31 cm[55] |
| DEXA | High precision and accuracy to differentiate total body fat, lean muscle, and bone<br>Safe for repeated measures | Differences between manufacturers and software versions<br>Unable to quantify muscle density | ALM < 20 kg for men, < 15 kg for women[56] |
| CT | Highly accurate quantitative and qualitative measurements<br>Useful in clinical settings where used as part of standard medical care | Large radiation exposure and costly if not obtained as part of routine care | Men < 52.4 $cm^2/m^2$<br>Women < 38.6 $cm^2/m^2$ [b,25] |
| BIA | Portable, safe, and inexpensive | Relies on population-specific regression equations and less accurate in altered states of hydration | Men < 10.75 $kg/m^2$<br>Women < 6.75 $kg/m^2$ [52] |
| MRI | Excellent-resolution images<br>Most accurate method to detect body composition at tissue-organ level<br>Safe | Costly and requires specialized personnel<br>Cannot accommodate very large patients | No consensus definition specific to this modality[20] |
| Ultrasound | Portable, safe, and inexpensive | Lacks standardized techniques causing significant interobserver variability<br>More qualitative than quantitative results<br>Requires specialized personnel | Can provide an estimate of lean muscle mass and apply same cutoffs as DEXA (see above)[57] |
| Deuterated creatine (methyl-$d_3$) | Accurate and reproducible measure of total muscle mass<br>Noninvasive and no radiation exposure | Needs further validation within oncology<br>Not yet accepted by guidelines<br>Requires urine collection on specified days after ingesting pill | $D_3Cr$ muscle mass/body mass < 0.273[58] |
| Patient-reported (such as SARC-F and SarcoPRO) | Easily obtained and inexpensive | Requires confirmatory testing as primarily screening tools<br>Not yet validated in oncology | SARC-F score of 4 or more[59] |

Abbreviations: ALM, appendicular lean mass; BIA, bioelectrical impedance analysis; CT, computed tomography; $D_3Cr$, $D_3$ creatine; DEXA, dual-energy x-ray absorptiometry; MRI, magnetic resonance imaging; SARC-F, strength, assistance with walking, rising from a chair, climbing stairs, and falls; SarcoPRO, sarcopenia patient-reported outcome.

[a]Definitions provided are those commonly used within the medical literature; however, many cut-points exist for several of these techniques and the correct cut-point is controversial. Depending on the context and scientific question, it may be more appropriate to use body composition measures as a continuous variable rather than depending on a dichotomized threshold.

[b]Many different CT-based definitions of sarcopenia exist and the one listed is the first proposed in the literature. See Table 1 for other commonly used definitions.

was associated with increased geriatric assessment impairments, reduced HRQOL, and inferior survival.[59] This tool and other patient-reported sarcopenia screening tools are attractive, given the ease with which they could be incorporated into clinical care, but require confirmatory testing afterward and warrant more validation in oncology. Finally, although rarely incorporated in oncologic studies and not available retrospectively, measuring muscle strength and physical performance is important in understanding sarcopenia.[60] Handgrip strength is a good and simple measure of muscle strength that correlates well with leg strength, poor mobility, and incident disability in activities of daily living (ADL).[61] A wide range of techniques are available for assessing physical performance, including short physical performance battery, usual gait speed, stair climb power test, and 6-minute walk test.[60] In older adult populations, physical performance assessments are often incorporated within geriatric assessments, which can be used to better understand sarcopenia and its consequences. As geriatric assessments are recommended as part of the routine care of older adults with cancer, this tool can be also be used to identify individuals at risk for sarcopenia.

The choice of measuring method of sarcopenia depends on several factors, including the specific purpose (clinical *v* research), the setting (inpatient *v* outpatient), and the cancer type (colorectal cancer *v* breast cancer, etc). There is no consensus regarding the best imaging modality to use to evaluate muscle mass at this time. In settings in which either CT, MRI, or DEXA imaging is used as part of routine care, these are preferred as they require no additional costs or radiation exposure. However, although these imaging

modalities may be archived and accessible, muscle mass is not regularly measured or reported on radiology reports. As the literature and importance of sarcopenia grows, hopefully, reporting this information will become more commonplace. Of note, as the majority of the oncology literature is based on cross-sectional CT derived cut-points of muscle mass, this is the definition used in the majority of the cited literature on associations with outcomes. Which CT derived cut-point to use in defining sarcopenia is an area of contention and depends on many factors. For research studies, we suggest using muscle measures as a continuous variable (often stratified by sex) rather than using a cut-point as dichotomizing muscle measures assumes that everyone below and above the cut-point is the same risk, which is most often not true. For clinical use, it is more convenient to use a cut-point, and choosing one derived from a similar population in terms of geography, ethnicity, cancer type, and disease status is important.

## SARCOPENIA AND ONCOLOGY OUTCOMES

In an umbrella review that included 30 meta-analyses of sarcopenia and adverse outcomes, sarcopenia was significantly associated with poorer prognosis across 12 cancer types: gastric, hepatocellular, urothelial, head and neck, hematologic malignancy, pancreatic, breast, colorectal, lung, esophageal, hematologic malignancies, and ovarian.[62] All studies examined associations with overall survival or all-cause mortality. Sarcopenia increases risk of recurrence and cancer-related death in gastric, urothelial, colorectal, and hepatocellular cancer. Importantly, almost all (95% CI) of the meta-analyses included in the review demonstrated that sarcopenia was related to poorer prognosis with overall hazard ratios (HRs) ranging from 1.11 to 2.12.[62] Greatest effects were seen in gastric, pancreatic, hepatocellular, and head and neck cancer. Similarly, in a recent meta-analysis of sarcopenia in malignant hematologic diseases, sarcopenia was an independent predictor of mortality disease (HR, 1.94).[63] Only a few meta-analyses have specifically examined the association of sarcopenia with mortality in older adults, but those studies were not limited to patients with a cancer diagnosis. In those studies, sarcopenia was associated with an increased risk of mortality with the magnitudes of effect similar to those seen in the aforementioned cancer studies.[64,65]

In the same umbrella review, five of the meta-analyses examined sarcopenia and 16 postoperative outcomes in GI cancers (all), esophageal, gastric, or colorectal cancer.[62] Sarcopenia was significantly related to one or more of major postoperative complications in each of those cancers. Where significant associations with sarcopenia and specific postoperative outcomes were found, effect sizes ranged from HRs of 1.35 to 6.24. Sarcopenia associated most strongly with postoperative pulmonary complications, infections, readmission rates, and length of hospitalization. Two additional systematic reviews, both in urologic cancers, had conflicting results. Sarcopenia was associated with a higher 90-day complication rate, whereas in the other, there was no association between sarcopenia and postoperative outcomes after urologic oncology surgery. In patients of age 65 years and older not restricted to cancer, low muscle mass was associated with increased length of hospital stay in endoscopic surgeries.[66] No reviews have specially addressed sarcopenia and surgical outcomes among older adults with cancer. Furthermore, sarcopenia was associated with complications and increased length of hospital stay in patients undergoing autologous or allogenic hematopoietic cell transplantation.[67,68]

In a systematic review done in 2016, Gerard et al[69] examined the association between body composition and chemotherapy or targeted therapy toxicity, which included 14 studies across the following cancer types: thyroid, colorectal, renal, hepatocellular, breast, esophageal, and ovarian cancer. Twelve of the 14 studies demonstrated higher chemotherapy toxicity associated with reduced lean mass or sarcopenia compared with normal or high lean mass. Of the various chemotherapy and targeted therapies studied, no significant association was reported between lean mass and chemotoxicity in studies of women receiving anthracyclines for breast and ovarian cancer. Evidence regarding the relationship between sarcopenia and chemotherapy toxicity specifically in older adults is lacking. Investigating this relationship in the geriatric oncology population is crucially important, given the known changes in loss of lean mass with age and the observation that approximately half of the older patients will present with severe toxicity.[70,71]

## EXPLAINING SARCOPENIA AND ADVERSE OUTCOMES

Although many studies have demonstrated the association of sarcopenia with adverse outcomes in cancer, only a few explanations have been proposed to explain these findings. Dosing of the majority of cytotoxic chemotherapeutics is based solely on height and weight and does not take into consideration differences in volume of distribution, drug metabolism, and clearance. Commonly, with aging, there is a decline in muscle mass while there is an increase in fat tissue. These body-composition changes result in the change of pharmacokinetics of both hydrophilic drugs (increasing the volume of distribution and extend the half-life time of hydrophilic drugs) and hydrophobic drugs (decrease in volume of distribution and also in case of hypoalbuminemia, increase free fraction bound of protein-bound drugs).[69,72] These body-composition changes result in the change of pharmacokinetics of both hydrophilic drugs (increasing the volume of distribution and extending the half-life) and hydrophobic drugs (decreasing the volume of distribution, and also in the case of hypoalbuminemia, increasing the free fraction of protein-bound drugs).[72]

Another proposed explanation is sarcopenia as a marker of an increased cancer-related inflammatory response. In a large-scale study of patients with colorectal cancer (n = 2,470) prediagnosis inflammation (measured as

neutrophil-to-lymphocyte ratio) was associated with at-diagnosis sarcopenia. Sarcopenia combined with inflammation nearly doubled risk of death from colorectal cancer (HR, 2.43; 95% CI, 1.79 to 3.29).[73] Chronic inflammation is characterized by increase in circulating proinflammatory cytokines, accompanied by elevated presence of dysfunctional T-regulatory cells and a T-cell–senescent phenotype. These changes decrease the immune response, resulting in a condition known as immune-senescence.[74] A study by Arrieta et al[75] evaluated the relationship of inflammatory markers and chemotherapy toxicity in patients with non–small-cell lung cancer who have received paclitaxel-cisplatin. Elevated inflammatory markers were associated with grade 3-4 toxicities as measured by platelet-lymphocyte ratio and anemia.

To date, there are limited prospective data on the added benefit of body composition beyond other known predictors of treatment toxicity and survival, such as geriatric assessment factors.[70] Critical prospective studies, including dedicated pharmacokinetics studies, are missing and highly needed to fully understand the relationship of sarcopenia with adverse outcomes and to inform effective and targeted interventions.

## SARCOPENIA IN SURVIVORSHIP

With the growing number of cancer survivors, there is increasing focus on the needs of cancer survivors, many of whom are older adults.[76] Many older adults with cancer have limitations in physical function and impairments in ADL and instrumental ADLs.[77,78] These impairments have been linked with sarcopenia and are likely related to worsening sarcopenia from a cancer diagnosis and cancer treatments. In a recent study of community-dwelling older adults, those diagnosed with cancer had an accelerated decline of appendicular lean muscle mass compared with noncancer and age-matched controls.[38] Figure 3 represents the hypothetical declines in sarcopenia associated with a cancer diagnosis. There is likely some acceleration of sarcopenia related directly to the cancer itself that occurs before diagnosis, which is likely further exacerbated by antineoplastic therapies. In patients who complete curative intent cancer treatments and enter the survivorship phase, it remains unknown whether there is recovery, continued accelerated declines, or rather a phase shift of sarcopenia (Fig 3). Given the rising number of older adult cancer survivors and functional independence being a high-priority outcome for the older population, future studies are needed to better understand sarcopenia as a survivorship issue amendable to targeted intervention.

## INTERVENING ON SARCOPENIA

Because of a high prevalence of sarcopenia among older adults, and its association with adverse outcomes, a number of interventions are being investigated to combat sarcopenia and minimize such adverse events. These include physical exercise, nutritional supplementation, hormone replacement, and therapeutic agents promoting muscle mass. However, most studies have been performed among noncancer patients, and studies focused among patients with cancer are relatively limited.[42,79]

Exercise interventions incorporating resistance training have been shown to increase both muscle mass and physical function among adults with cancer. In a cohort of patients with metastatic prostate cancer receiving androgen deprivation therapy, where muscle loss is common, a combined resistance and aerobic exercise program helped combat and reverse the loss of muscle.[80] Another study testing 12 weeks of resistance training versus usual care among 37 patients with prostate cancer receiving androgen deprivation therapy similarly reported improvement in lean mass, sarcopenia prevalence, muscle strength, and HRQOL.[81] Another study among patients with breast cancer receiving adjuvant chemotherapy randomized resistance exercise training (n = 64), aerobic exercise training (n = 66), or usual care (n = 70) demonstrated that resistance exercise can reverse sarcopenia and improve quality of life.[82] In a meta-analysis of 11 randomized controlled trials examining the impact of resistance training in cancer survivors of all ages, resistance training was shown to be associated with improvements in muscle function and body composition.[83] Nonetheless, major barriers exist to implementing these interventions as many community-dwelling older adults lack access or motivation to partake in a rigorous exercise program and the optimal type, intensity, and timing of exercise interventions remain unknown.[83,84]

Malnutrition is highly prevalent among older adults with cancer and represents one of the mechanisms contributing to sarcopenia.[85,86] Although the recommended protein requirement for older individuals (0.8 g/kg per day) has remained unchanged for decades, a higher protein intake may be needed to optimize muscle protein synthesis.[87] Aside from physical activity status, total daily protein intake appears to be the most important determinant of skeletal muscle mass, and several studies have examined acute and chronic effects of nutritional supplementation among older adults.[88] In a study of 61 patients undergoing cystectomy for bladder cancer randomly assigned to oral nutritional supplementation versus multivitamin supplementation, a reduced prevalence of sarcopenia and fewer adverse events or readmissions were seen favoring the oral nutrition group.[89] Another study randomly assigned patients with colorectal cancer undergoing chemotherapy into whey protein supplementation versus placebo and reported improved sarcopenia indices and nutritional status favoring the intervention arm.[90] However, in a large individual patient data meta-analysis of eight randomized trials incorporating 557 older adults studying the effect of protein or

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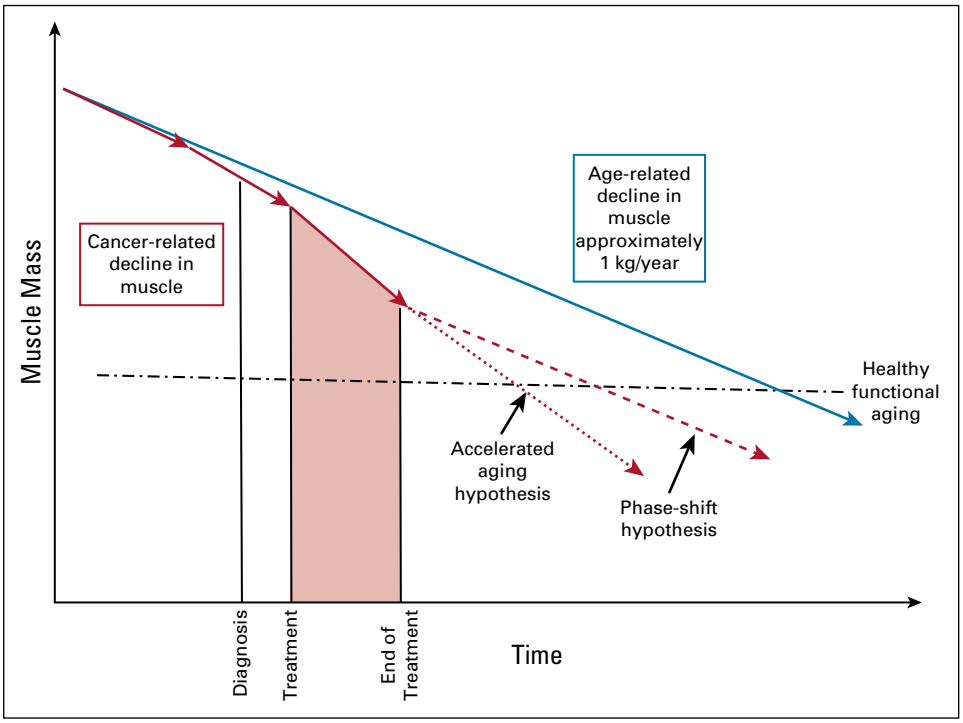


**FIG 3.** Hypothetical graph of the changes in muscle mass in older adults before and after a cancer diagnosis in comparison to normal age-related declines.

amino acid supplementation on muscle mass and strength, no significant positive effects of protein or amino acid supplementation on muscle mass or strength were seen.[91] Combinations of exercise and nutritional supplementation may have an additive effect on optimizing muscle protein synthesis and reversing sarcopenia, and trials are ongoing to test the effectiveness of such multimodal interventions among older adults with cancer.[92]

Finally, several pharmacologic interventions for combating sarcopenia remain under investigation. Studied agents include antibodies against myostatin (eg, stamulumab and trevogrumab), activin receptor agonists (eg, bimagrumab), exercise mimetics (eg, PPARβ/δ agonist), and selective androgen receptor modulators (eg, enobosarm). However, a clear benefit from larger definitive phase III trials remains to be seen and as a result, currently no US Food and Drug Administration–approved therapeutic agents exist for treatment of sarcopenia.[79]

## DISCUSSION

The growing field of body composition in cancer holds great promise in better assessing and treating older adults with cancer, yet the research of body composition and geriatric oncology has many unanswered questions. How is the mechanism or process of muscle loss different between aging patients with and without cancer? How should sarcopenia be defined in oncology? What factors and interventions modify the adverse trajectories of sarcopenia related to cancer and cancer treatments? How can we best treat and help to maintain muscle mass and function? The care of older adults with cancer is challenging with the dual focus to maintain the quality of life and functional independence as best as possible while at the same time maintaining treatment efficacy. The use of body composition in assessing older adults with cancer is a promising avenue to provide more personalized oncologic treatment that could ultimately improve outcomes.

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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

**Sarcopenia in the Older Adult With Cancer**

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